Dear Editor,

Please consider our manuscript "Silent but deadly? Physics says seldom, memory says mostly" for publication as a Research Article in *Journal of the Royal Society Interface*.

The belief that a silent release of flatus smells worse than an audible one is recorded in many languages, yet no mechanistic test of it exists. The closest prior work is a conference abstract on the physics of flatulence (Ancalle et al. 2021). We follow one passage through a single chain: chemistry, sound, dispersion, perception and memory. Three results reach beyond the topic.

1. **Routing by the human convective boundary layer.** In simulations of a sofa, a standing pair and a closed lift, the warm air rising along a body takes a slow release next to it and lets a fast one pass, so the momentum of the release, relative to that of the boundary layer, decides which nearby nose it reaches. The same set-up reproduces measured breathing-zone intake from a source at a seated manikin's groin within a factor of 1.5. This bears on any near-body source, such as respiratory, skin and clothing emissions in shared indoor spaces.

2. **A peak bound for linear transport.** We prove that a slower release of the same substance cannot give a higher peak concentration while both are carried passively by the same flow, whatever that flow is. A slower release can win only near the source: through less pre-dilution, by at most the jet-to-ambient velocity ratio (which we call the Silencer number), or by being carried to a different nose.

3. **Selection across two senses.** With passages as emitted, a bystander on a sofa, where people spend most of their shared time, finds the silent passage worse in 0.5% of pairs and the audible one in 9.2%; only in a closed lift, for the person behind the releaser, is the silent one worse as often. Yet because a silent passage is noticed only if it is smelled, a remembered silent passage seems the worse one in 88% of comparisons. When an event can be noticed through either of two senses, selection on noticing alone can create a false sensory belief. The explanation makes testable predictions: releasers should hold the belief less firmly than bystanders, and people without a sense of smell should never remember a silent passage.

The work joins fluid dynamics, chemistry, acoustics, sensory science and cognition. We think it suits *Interface* because each part can be read on its own, and because the question shows how a physical and a cognitive explanation can be separated. The outcomes are reported per setting and checked against the scene weights, the observer models, large-eddy peak factors, mesh refinement and clothing. On the sofa the order holds with the numerical uncertainties combined at their median values, and the remembered share stays above 60% in every case.

Because the work spans several fields, our suggested reviewers cover them: indoor plume and ventilation modelling (H. C. Burridge), the human thermal plume and CFD validation (I. Năstase), the human convective boundary layer and breathing-zone exposure (D. Licina, whose manikin data we use for validation), CFD of inhaled and smelled pollutants near the body (K. Ito), turbulent puffs and jets (S. S. Diwan), olfactory perception (A. Oleszkiewicz) and judgement from selective samples (H. Alves). We have no collaboration with any of them.

Proofs and technical details are in the electronic supplementary material. Code and processed data are openly available at `https://github.com/ccy5123/sbd`; an archived release with a DOI (Zenodo) will accompany the revised manuscript. A preprint is posted on engrXiv (`https://doi.org/10.31224/8708`). The results have not been reported in the media. The manuscript is not under consideration elsewhere. AI-assisted technologies (Claude Code, Anthropic) were

used to write code and simulation set-ups, search the literature and draft text, as declared in the manuscript.

Yours sincerely,

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